



Introduction to Earth System

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Department of Space and Applications

1

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1



Evaluation and Scoring

Component	Attendance	Exercises	Practical	Reports	Midterm	Final
Percentage %	10	20	20	0	0	50

2

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2



Further reading

1. Wallace J.M and P.V. Hobbs: **Atmospheric Science, An Introductory Survey**, 2nd Edition, Academic Press, 504pp. [[PDF](#)]
2. IPCC assessment reports (TAR, AR4, AR5, AR6) (<https://ipcc.ch>)

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3

3



Course content

- I. Introduction
- II. Earth's components
- III. Daisy world
- IV. The radiation balance of Earth
- V. Large-scale oscillation & climate change

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4

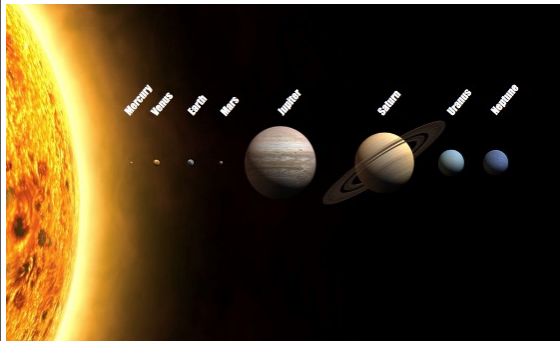
4



I. Introduction

Our Earth is:

- 'one point' in the universe
- Age of the universe?
- Why Earth is the only planet presently known to support life?



- Age of the Earth?
- Orbital period?
- Mean radius?

Distance from the Earth to the Sun: 1 Astronomical Unit
(1 AU: 150.000.000 km)

(distance not to scale)

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5

5



Orbits

■ Ptolemy (AD90–AD168, Greek)

- geocentric model



Ptolemy

■ Nicolas Copernicus (1473–1543)

- heliocentric model
 - the Earth circles the Sun
 - the planetary orbit is a circle



Copernicus

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6

6

Orbits

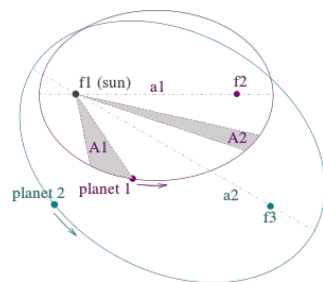
■ Johannes Kepler (1571–1630)

3 Kepler's laws of planetary motion

1. The orbit of a planet is an ellipse with the Sun at one of the two foci (1609)
2. A line segment joining a planet and the Sun sweeps out equal areas during equal intervals of time (1609)
3. The square of the orbital period of a planet is proportional to the cube of the semi-major axis of its orbit (1619)



Orbits



3 Kepler's laws of planetary motion

$$\frac{a_1^3}{T_1^2} = \frac{a_2^3}{T_2^2} = \dots = \frac{a_i^3}{T_i^2} = \dots = K$$

$$K = G (m_p + m_s) / (4 \pi^2)$$

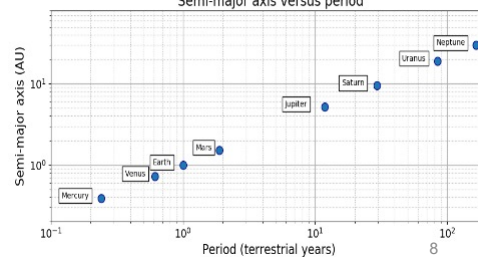
G: gravitational constant

m_s : mass of the sun

m_p : mass of the planet

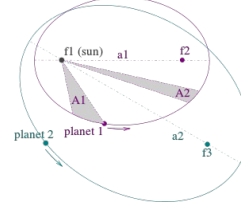
$$G = 6.674 \times 10^{-11} \text{ N} \cdot (\text{m/kg})^2$$

Solar System planets
Semi-major axis versus period





Orbits



Q1. Find the common orbital characteristics of the Moon, Hubble telescope, and the ISS!

Q2. Is the speed of planets/ satellites constant along their orbits?

Q3. Consider the Geostationary Orbit (GEO). A satellite in GEO has an orbital period exactly equal to the Earth's rotation period. If the Earth's rotation rate were to slow down 1% due to atmospheric drag or internal redistribution of mass, what would happen to the required altitude (a) of a satellite to remain in a geostationary position?

9

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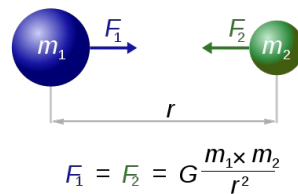
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Orbits

Newton's law of universal gravitation

- G is the gravitational constant ($6.674 \times 10^{-11} \text{ N} \cdot (\text{m}/\text{kg})^2$)



Q4. At what velocity will the satellite not fall back to the Earth?

Q5. At what velocity can a satellite escape from the gravitational attraction of the Earth?

Q6. At what velocity can a satellite escape from the solar system?

Q7. At what speed does the Earth orbit the Sun?

Earth-Sun distance: 1 AU~150.000.000 km

10

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10



Orbits

Balance between gravitational force and centrifugal force

$$mg = \frac{mv^2}{R} \rightarrow v = \sqrt{gR}$$

where $g = \frac{GM}{R^2}$

$g \sim 9.8 \text{ m/s}^2$, $R \sim 6378 \text{ km} \rightarrow v_1 \sim 7.9 \text{ km/s}$

$v_3 \sim 16.6 \text{ km/s}$

velocity that the satellite can escape from the solar system

Conservation of Energy (kinetic energy & gravitational potential energy)

$$\frac{mv_2^2}{2} - \frac{GMm}{R} = 0$$

$$\rightarrow v_2 = \sqrt{2}v_1 \rightarrow v_2 \sim 11.2 \text{ km/s}$$

Escape velocity: The minimum speed needed for an object to escape from the gravitational attraction of a massive body

- Geostationary orbit
- Polar orbit
- Low orbit

- When $v_1 < v < v_2 \rightarrow$ The satellite has an elliptical orbit around the Earth
- When $v_2 < v < v_3 \rightarrow$ become a satellite/planet of the Sun

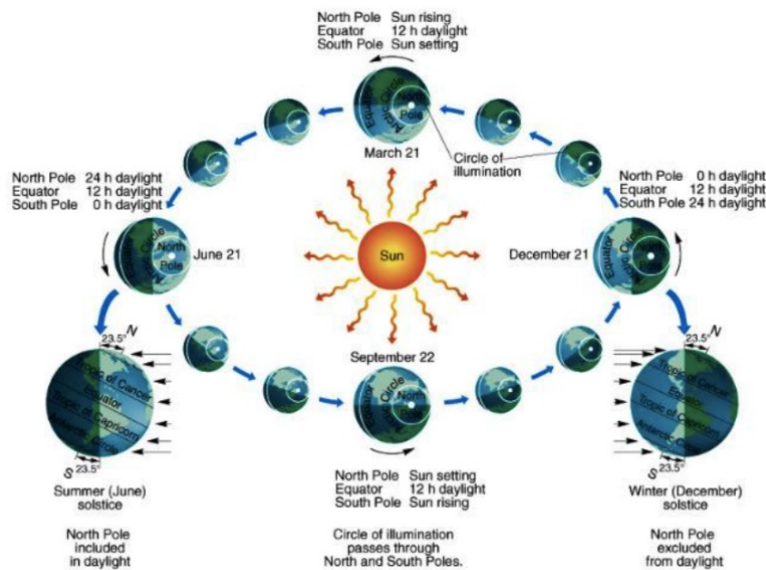
11

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11



Milankovitch Theory: The 3 Milankovitch Cycles

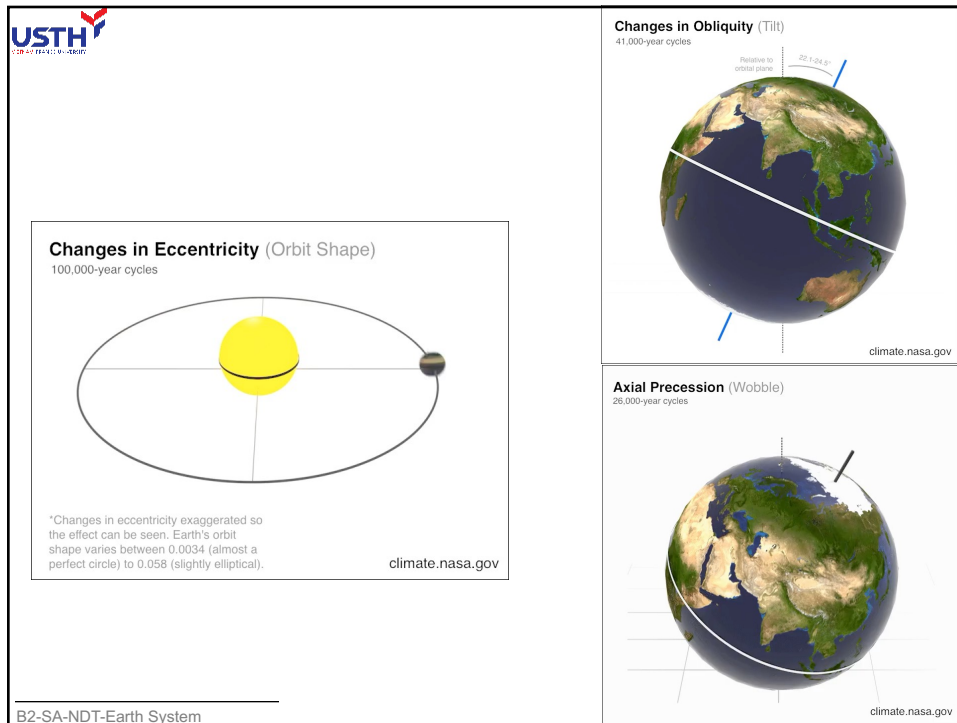


Earth's rotation and Revolution

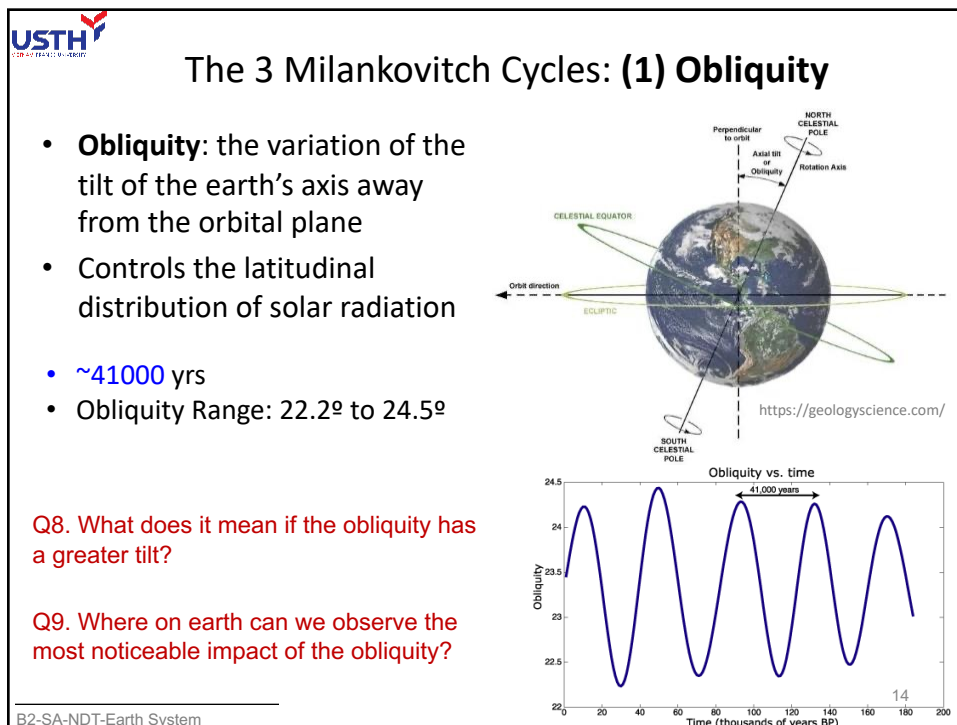
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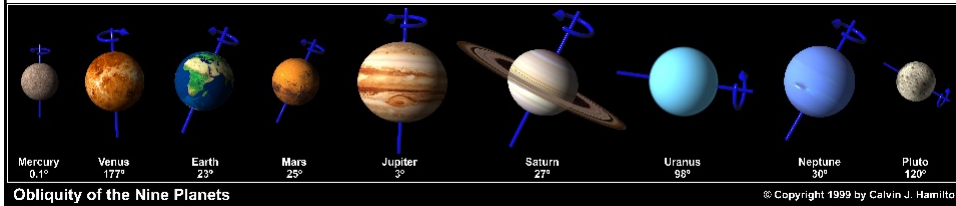
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14



Obliquity of the Planets, and the direction of spin



Q: Identify pairs of planets with similar obliquity?

Q: Which planet has the most peculiar spin direction?

15

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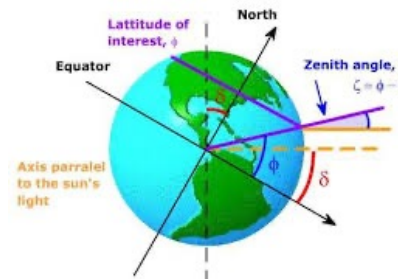
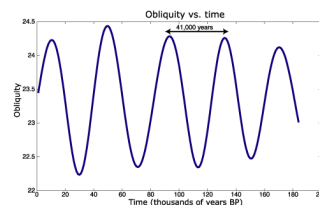


(1) Obliquity

Q10.

- Calculate the difference in the solar zenith angle at the North Pole (90°N) when Earth is at minimum obliquity (22.1°) versus maximum obliquity (24.5°).
- How does this change the "intensity" of the sun at the pole?

Hint: the intensity of solar radiation (I) received at the surface is proportional to the cosine of the zenith angle



16

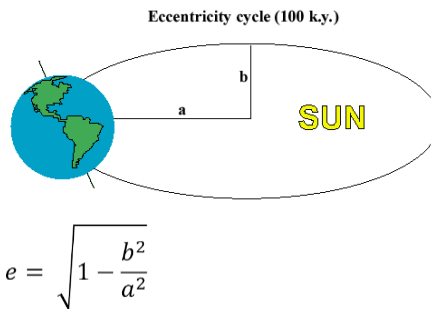
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16



The 3 Milankovitch Cycles: (2) Eccentricity

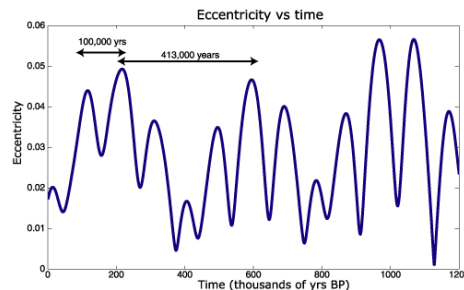
Eccentricity: indicates the shape of Earth's orbit around the sun



Eccentricity:

- -0.005 to 0.06 (0 → a circle)
- currently 0.017
- ~ less than 0.2% change in solar insolation, equivalent to a **radiative forcing** of ~0.45 W/m²

- Period: **100,000** dominates and the **413,000** amplifies every 4th one



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17

17



$$e = \sqrt{1 - \frac{b^2}{a^2}}$$

(2) Eccentricity

Q11. Estimate the perihelion distance r_p (via a : semi-axis & e)

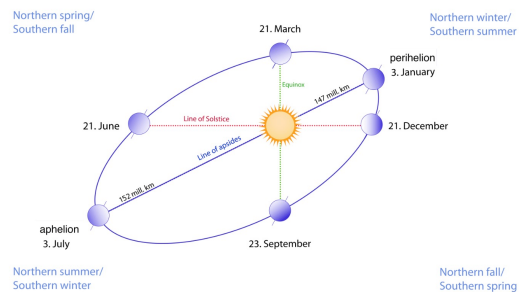
Q12. Estimate the aphelion distance r_a (via a : semi-axis & e)

Q13. $e=0.017$

What is the ratio of the solar flux received at perihelion (S_p) versus aphelion (S_a)?

Q14. $e=0.06 \rightarrow ? S_p/S_a$

Q15. If the Northern Hemisphere summer currently occurs near aphelion. Would a higher eccentricity ($e=0.06$) make the NH summers warmer or cooler?



"cool summers" at high northern latitudes → snow might not be melt completely → snow accumulation → increase Earth's albedo → triggering a **positive feedback** loop → **potential initiation of an ice age**

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18

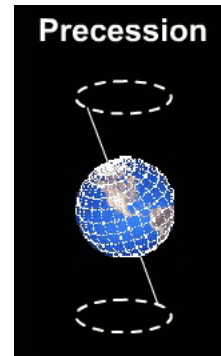
18



The 3 Milankovitch Cycles: (3) Precession

Precession: change in orientation of the Earth's rotational axis; ~26,000 years

- Precession influences the timing of the seasons
 - Currently, the NH's Summer Solstice occurs near **Aphelion**
 - Precession → the timing of these seasonal events drifts slowly around the orbit.
 - In ~13,000 years, the precession cycle will have rotated the Earth's axis by 180° → the NH's Summer Solstice will occur at **Perihelion** → **Amplification of Seasonal Extremes.**
- Precession greatly impacts on Glacial cycles
- Precession fundamentally changes how we view the "night" sky
 - Our North Pole currently points to Polaris
 - In 13000 yrs → it will point toward Vega

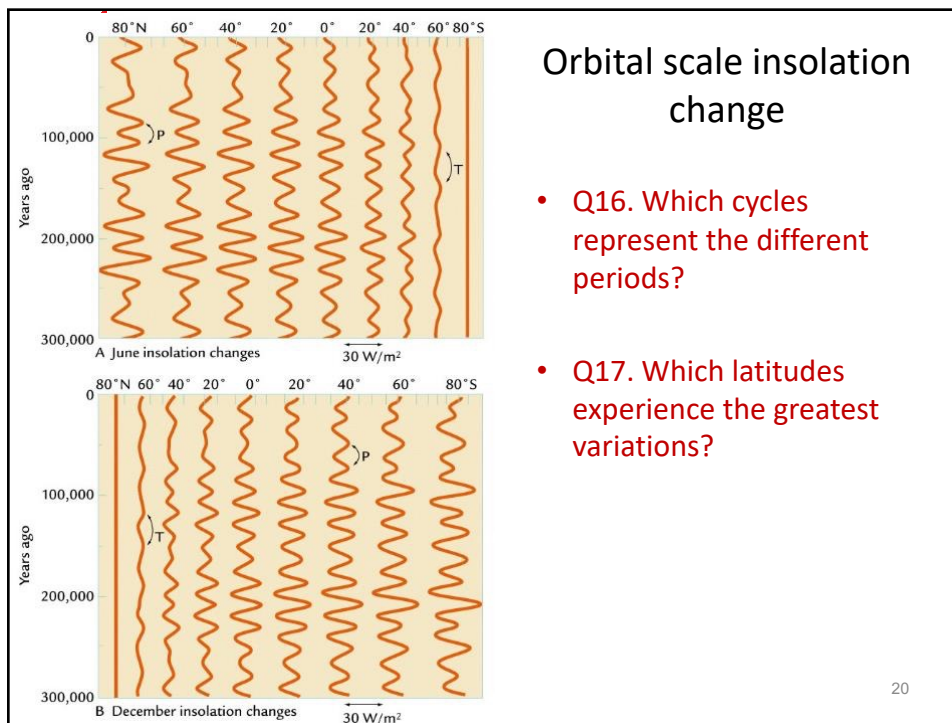


rotation of the Earth's rotational axis

19

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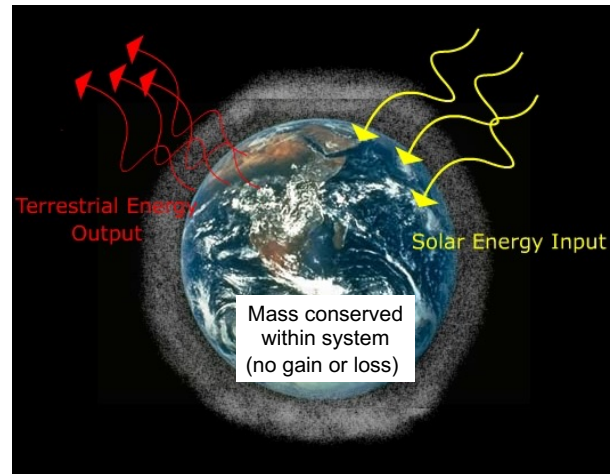
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The Earth

- exchange of energy but negligible exchange of mass with surroundings



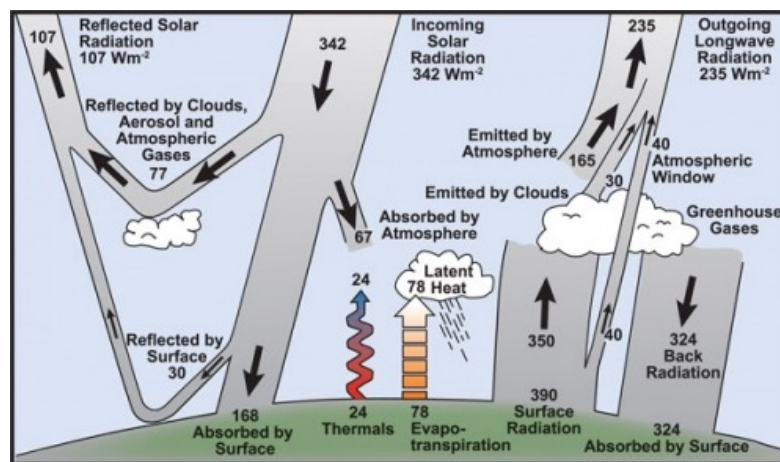
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21

21



Energy balance



- When the climate system reaches steady state, i.e. temperature stops changing
→ amount of energy going out = amount of energy coming in
- **Radiative forcing**

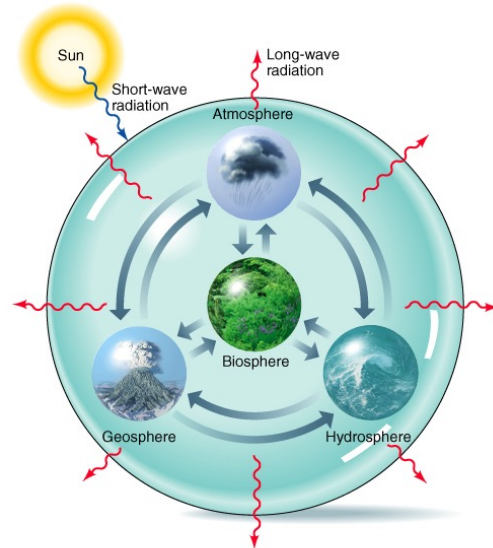
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22

22

Four Spheres of the Earth System

- **A system:** is a group of components that interact



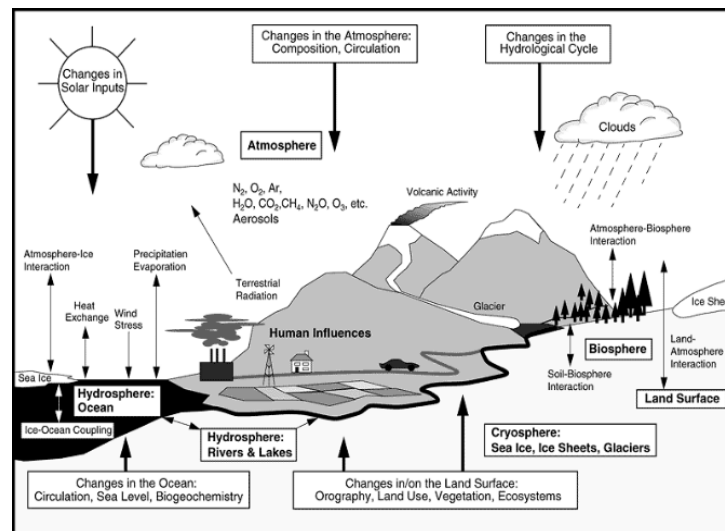
We focus more on the Atmosphere in this course

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23

23

Components of the Earth System



IPCC, 2001 24

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24

Temporal scales

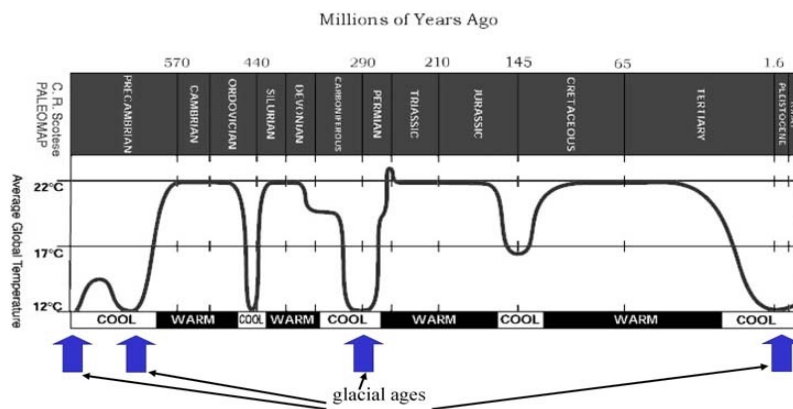
- **Millions of years:** tectonic, species distributions, etc.
- **Thousands years:** evolution of Earth's orbit
- **Decades to centuries:** recent global warming
- **Days/seasons:** weather, plant growth
- **Day:** heating & cooling, volcanoes, earthquakes, etc.

25

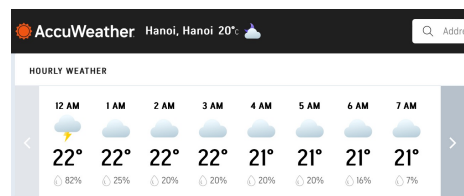
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25

Temporal scales



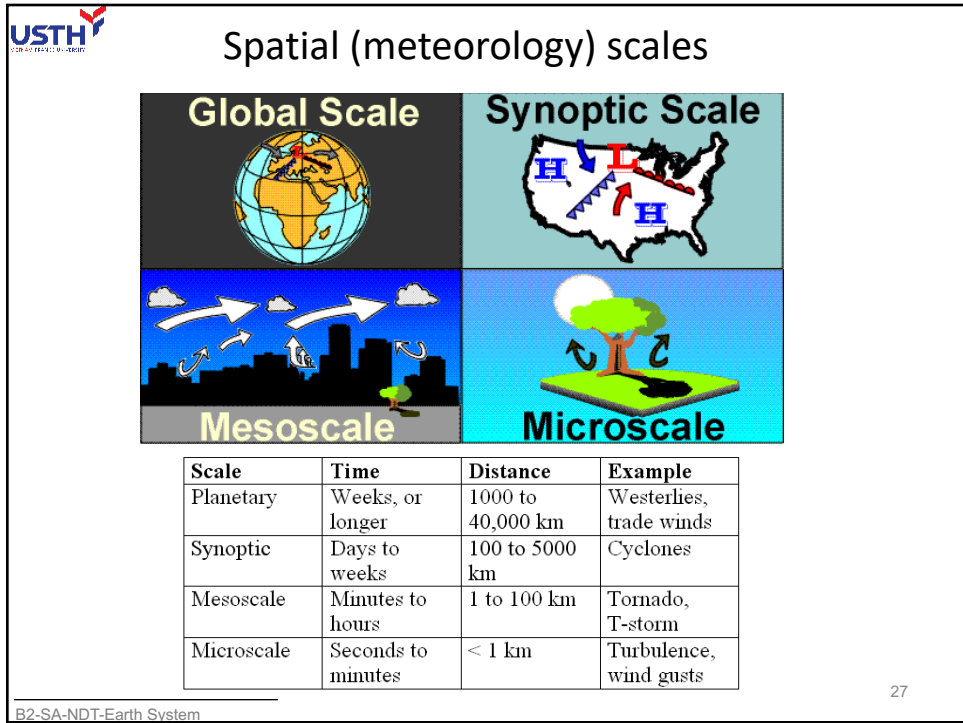
Hourly – daily weather



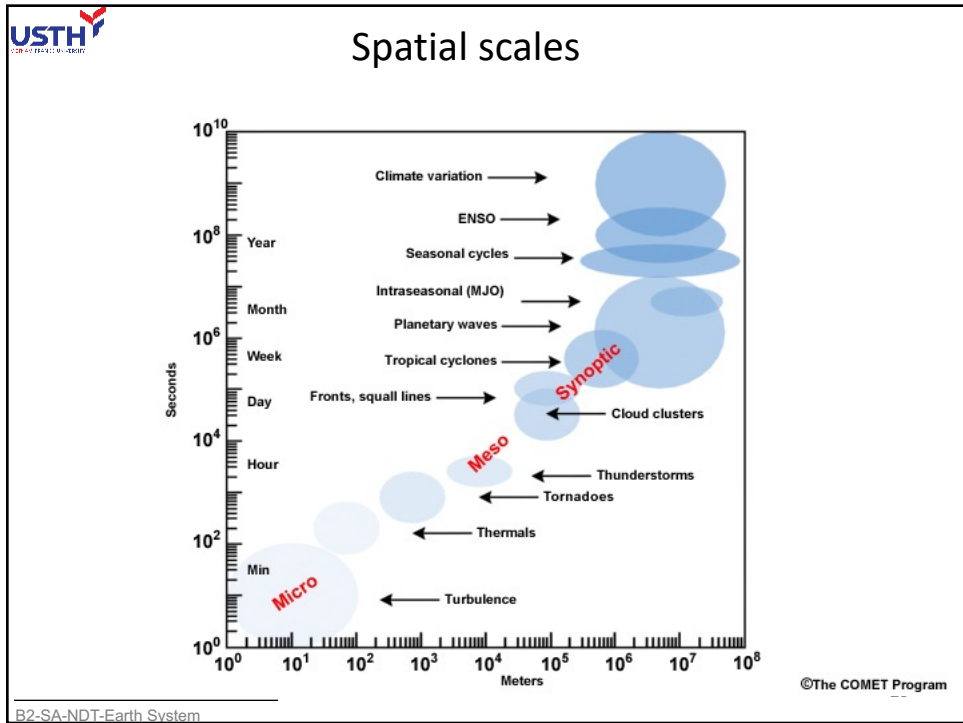
26

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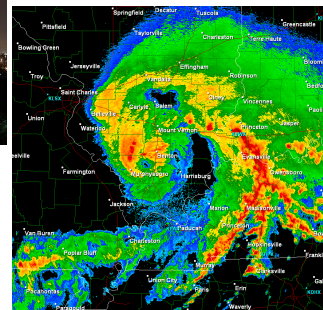
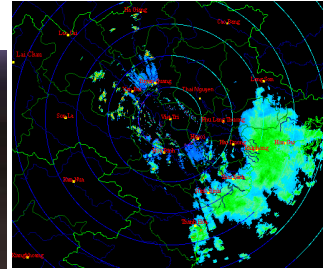


27





▪ Mesoscale



- e.g. cumulus, showers, thunderstorms, tornadoes (between microscale & mesoscale)

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31



▪ Microscale



300m



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32

32



Quiz

Which of the following weather phenomena is typically classified as a microscale event?

- A. Sea breezes
- B. Mid-latitude cyclones
- C. Tornadoes
- D. Jet streams

33

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33



Quiz

What is the typical horizontal spatial range of mesoscale weather phenomena?

- A. Less than 1 km
- B. 5000 km to 40000 km
- C. 1 km to 100 km
- D. 100 km to 5000 km

34

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34



Quiz

Which scale is most associated with the daily weather maps seen on television?

- A. Synoptic scale
- B. Microscale
- C. Global scale
- D. Mesoscale

35

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35



Quiz

What is a primary characteristic of a microscale weather phenomenon's lifespan?

- A. Several days
- B. Several hours
- C. Seconds to minutes
- D. Several weeks

36

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36



Quiz

If you are tracking a localized urban heat island effect, which scale are you most likely investigating?

- A. Synoptic scale
- B. Microscale
- C. Global scale
- D. Planetary scale

37

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37



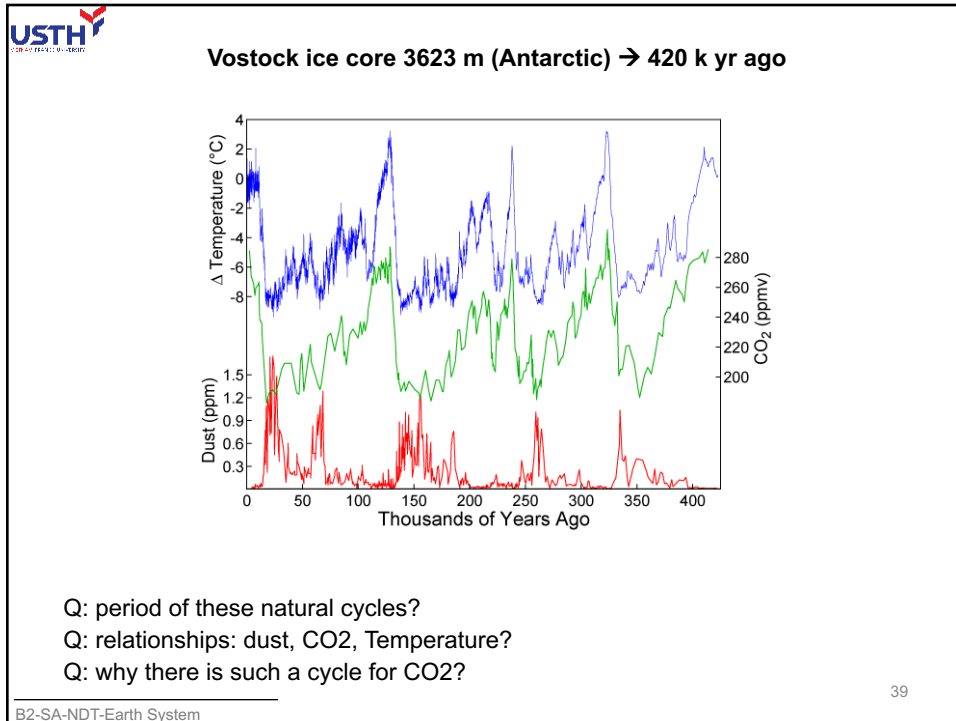
Summary

1. Earth: general characteristics & orbit
2. Milankovitch theory
3. 4 components of the Earth
4. Temporal & spatial scales

38

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39

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Homework 1

- Download the Vostock Ice Core data:
 - <https://www.ncei.noaa.gov/pub/data/paleo/icecore/antarctica/vostok/deutnat.txt>
 - <https://www.ncei.noaa.gov/pub/data/paleo/icecore/antarctica/vostok/co2nat.txt>
 - <https://www.ncei.noaa.gov/pub/data/paleo/icecore/antarctica/vostok/ch4nat.txt>
 - <https://www.ncei.noaa.gov/pub/data/paleo/icecore/antarctica/vostok/dustnat.txt>
- Where is Vostock, and why is this station used for ice core sampling?
- Plot the figures showing temporal evolutions of CO₂, CH₄, temperature, and dust
- Comment on the relationships between these variables and highlight and periods

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40

40



Practice #1: Preparation for Practical session using Python

Install Anaconda & Visual Studio

- Install anaconda/miniconda
<https://www.anaconda.com/products/individual>
- Install Visual Studio Code
<https://code.visualstudio.com>
- Go through:
<https://code.visualstudio.com/docs/getstarted/getting-started>